AI Playbook and Toolkit for Public Health Departments

AI Workflow Standard Operating Procedures

OPS 230

An AI workflow standard operating procedure translates AI policy into day-to-day practice. It describes how staff should use an approved AI-supported workflow, what they are responsible for reviewing, what they must document, and when they must escalate a concern. In public health, SOPs are especially important because AI outputs may influence surveillance, communications, case review, service referrals, resource allocation, or internal decision support. This module teaches learners how to create practical SOPs for AI-supported workflows. The emphasis is on operational clarity. Staff should not have to infer whether they may paste data into a tool, accept a summary, override a recommendation, save an output, or report an error. The SOP should make those expectations explicit. AI SOPs help agencies move from principles to reliable implementation. They reduce variation across staff, support training and onboarding, create evidence for audit and incident review, and make it easier to monitor whether AI is being used as intended.

Level	applied/foundational
Primary track	Operations and Data Quality Track
Appears in tracks	operations-and-data-quality, program-management, governance-and-security, epidemiology

Learning Objectives

- Explain why approved AI tools still require workflow-specific SOPs.
- Identify the core elements of an AI workflow SOP.
- Define human review, documentation, override, escalation, and fallback expectations.
- Distinguish between acceptable and prohibited uses within a specific AI workflow.
- Produce a draft AI workflow SOP for a public health use case.

Module Overview

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This module teaches learners how to create practical SOPs for AI-supported workflows. The emphasis is on operational clarity. Staff should not have to infer whether they may paste data into a tool, accept a summary, override a recommendation, save an output, or report an error. The SOP should make those expectations explicit.

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Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Public Health Example

A health department approves an AI tool to draft summaries of environmental health complaints. The SOP explains which complaint types are eligible, which data fields may be used, how staff must compare the draft to the original complaint, where the final summary is stored, how unsupported statements are removed, and when errors must be reported. The SOP also explains that the tool may not issue inspection decisions, enforcement recommendations, or public notices without human review.

Technical and Operational Deep Dive

An AI SOP should begin with scope. The scope explains which workflow the SOP covers, which staff roles may use the tool, which systems or data are involved, and which AI-supported tasks are allowed. Scope is important because many AI tools can technically perform more tasks than the agency has approved. A clear scope prevents an approved tool from drifting into unapproved uses.

The SOP should define the standard workflow step by step. This includes preparing data, using the approved tool, reviewing outputs, correcting errors, documenting the final decision, and escalating concerns. For generative AI, the SOP should require staff to verify outputs against source material. For predictive tools, it should require staff to interpret scores in context and avoid treating the output as an automatic decision.

A strong SOP also includes exception handling. Staff need to know what to do when the AI tool is unavailable, the output is inconsistent with source data, the output appears biased or unsafe, the workflow involves a sensitive or unusual case, or the user is uncertain whether the use is allowed. The SOP should identify a fallback process and escalation pathway so staff are not forced to improvise.

SOPs should be maintained as living documents. AI workflows may change when policies, data feeds, models, prompts, vendors, or public health conditions change. SOP ownership, review frequency, and change control should be documented so the procedure remains aligned with approved use.

Risks, Failure Modes, and Guardrails

Risks and failure modes include the following:

Staff applying an AI tool beyond the approved workflow.

Outputs being copied into official records without required review.

Errors, bias concerns, or privacy issues not being escalated.

No fallback process when the tool fails or is unavailable.

Outdated SOPs that no longer match the deployed workflow.

Recommended guardrails include the following:

Create a workflow-specific SOP for each approved AI use.

Define allowed uses, prohibited uses, human review, and documentation requirements.

Require staff training before independent use.

Include override, escalation, and fallback procedures.

Review SOPs after incidents, major updates, or policy changes.

Reflection Questions

Where does this topic appear in your agency, program, or workflow?

Which staff roles would need to understand or apply this concept before AI use is approved?

What could go wrong if this topic is not addressed before implementation?

What evidence would governance, leadership, or operations staff need before moving forward?

Which policy, data, equity, privacy, security, or workflow questions remain unresolved?

Practical Exercise

Draft an AI workflow SOP for one approved or realistic public health use case. The SOP should include purpose, scope, roles, allowed uses, prohibited uses, step-by-step workflow, human review requirements, documentation requirements, escalation triggers, fallback process, and review schedule.

Expected Artifact or Evidence

Draft AI workflow SOP

Role and responsibility matrix

Allowed/prohibited use section

Escalation and fallback process

Review and maintenance schedule

Expected Assignment

- Draft AI workflow SOP
- Role and responsibility matrix
- Allowed/prohibited use section
- Escalation and fallback process
- Review and maintenance schedule

Knowledge Check

- 1. Why does an approved AI tool still need a workflow-specific SOP?
- 2. What is an override?
- 3. Which item should an AI SOP include?
- 4. When should an SOP be reviewed?
- 5. What is strong completion evidence?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP guidance for LLM application security
- Agency policies for privacy, cybersecurity, records, procurement, accessibility, and public communications
- Relevant public health program SOPs, data governance documentation, and implementation plans